Continuum Mechanics

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Chapter 01

Tensor Algebra

Part 3 of 5

Special Classes of Tensors

Transpose, Symmetry, and Positive Definiteness

Definition

For $S \in \mathcal{V}^2$, the *transpose*, or *adjoint*, $S^T \in \mathcal{V}^2$ is defined by

$$\mathbf{v} \cdot (S\mathbf{u}) = \mathbf{u} \cdot (S^T\mathbf{v}), \qquad \mathbf{u}, \mathbf{v} \in \mathcal{V}.$$

The tensor S is *symmetric*, or *self-adjoint*, if $S^T = S$, and *skew-symmetric* if $S^T = -S$.

Definition

A tensor S is *positive definite* if

$$\mathbf{v} \cdot (S\mathbf{v}) > 0 \qquad \text{for every } \mathbf{v} \neq \mathbf{0}.$$

It is *symmetric positive definite (SPD)* if it is both symmetric and positive definite.

Exercise

Show that the transpose is uniquely defined: if $A, B \in \mathcal{V}^2$ *both satisfy*

$$\mathbf{v} \cdot (S\mathbf{u}) = \mathbf{u} \cdot (A\mathbf{v}) = \mathbf{u} \cdot (B\mathbf{v}) \qquad \textit{for all } \mathbf{u}, \mathbf{v} \in \mathcal{V},$$

then $A = B$.

Invertible Tensors

Definition

A tensor $S \in \mathcal{V}^2$ is *invertible* if there exists $A \in \mathcal{V}^2$ such that

$$SA = AS = I.$$

In this case, we write $S^{-1} = A$.

Proposition

If S is invertible, its inverse is unique. Moreover, S^T is invertible and

$$(S^T)^{-1} = (S^{-1})^T.$$

Proof.

Exercise. $\square$

Remark

We use the unambiguous notation

$$S^{-T} := (S^T)^{-1} = (S^{-1})^T.$$

Definition

A tensor $Q \in \mathcal{V}^2$ is *orthogonal* if

$$Q^T = Q^{-1}.$$

Equivalently,

$$QQ^T = Q^T Q = I.$$

Rotations

Definition

An orthogonal tensor $Q \in \mathcal{V}^2$ is called a *rotation* if, for every right-handed orthonormal frame

$$\mathcal{F} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\},$$

the set

$$\mathcal{G} = \{Q\mathbf{e}_1, Q\mathbf{e}_2, Q\mathbf{e}_3\}$$

is also a right-handed orthonormal frame.

Orthogonality preserves the dot-product frame rule, while the additional right-handed condition preserves the displayed cross-product frame rules.

Tensor Operations in a Frame

Proposition

Suppose $S \in \mathcal{V}^2$ and $\mathcal{F} = \{\mathbf{e}_i\}$ is a frame. If $[S]$ denotes the matrix representation of S in $\mathcal{F}$, then

$$[S^T] = [S]^T.$$

If S is invertible, then

$$[S^{-1}] = [S]^{-1}, \qquad [S^{-T}] = [S]^{-T}.$$

Result 1.2 (Symmetric–Skew Decomposition)

For every $S \in \mathcal{V}^2$, there exist unique tensors $E, W \in \mathcal{V}^2$ such that

$$S = E + W, \qquad E^T = E, \qquad W^T = -W.$$

Proof.

Set

$$E := \frac{1}{2}(S + S^T), \qquad W := \frac{1}{2}(S - S^T).$$

Then $S = E + W$. It remains to establish symmetry, skew-symmetry, and uniqueness.

Proof (Cont.)

We use the elementary transpose identities

$$
\begin{aligned}
(A+B)^T &= A^T + B^T, && A, B \in \mathcal{V}^2, \\
(\alpha A)^T &= \alpha A^T, && A \in \mathcal{V}^2, \quad \alpha \in \mathbb{R}, \\
(A^T)^T &= A, && A \in \mathcal{V}^2.
\end{aligned}
$$

Therefore,

$$E^T = \frac{1}{2}(S^T + S) = E$$

and

$$W^T = \frac{1}{2}(S^T - S) = -W.$$

Proof (Cont.)

For uniqueness, suppose

$$S = E_1 + W_1 = E_2 + W_2,$$

where E_1, E_2 are symmetric and W_1, W_2 are skew-symmetric. Then

$$\boldsymbol{\Delta} := E_1 - E_2 = W_2 - W_1.$$

Thus $\boldsymbol{\Delta}$ is both symmetric and skew-symmetric. Hence

$$\boldsymbol{\Delta}^T = \boldsymbol{\Delta} \qquad \text{and} \qquad \boldsymbol{\Delta}^T = -\boldsymbol{\Delta},$$

so $\boldsymbol{\Delta} = O$. Therefore $E_1 = E_2$ and $W_1 = W_2$. $\square$

Example (Computing the Symmetric and Skew Parts)

Let

$$[S] = \begin{bmatrix} 2 & -1 & 4 \\ 3 & 0 & 5 \\ -2 & 1 & 6 \end{bmatrix}.$$

Then

$$[S]^T = \begin{bmatrix} 2 & 3 & -2 \\ -1 & 0 & 1 \\ 4 & 5 & 6 \end{bmatrix}.$$

We compute

$$[E] = \frac{1}{2}\left([S] + [S]^T\right), \qquad [W] = \frac{1}{2}\left([S] - [S]^T\right).$$

Example (Computing the Symmetric and Skew Parts (Cont.))

Carrying out the entry-by-entry calculations gives

$$[E] = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & 3 \\ 1 & 3 & 6 \end{bmatrix}, \qquad [W] = \begin{bmatrix} 0 & -2 & 3 \\ 2 & 0 & 2 \\ -3 & -2 & 0 \end{bmatrix}.$$

Directly,

$$[E]^T = [E], \qquad [W]^T = -[W], \qquad [E] + [W] = [S].$$

Thus the symmetric part retains the entries mirrored across the diagonal, while the skew part records their antisymmetric differences.

Cross-Product Rules Used Below

The calculation in Result 1.3 does not require a geometric definition of the cross product. We use its linearity in each argument and its action on a right-handed orthonormal frame:

$$\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3, \qquad \mathbf{e}_2 \times \mathbf{e}_3 = \mathbf{e}_1, \qquad \mathbf{e}_3 \times \mathbf{e}_1 = \mathbf{e}_2,$$

$$\mathbf{e}_2 \times \mathbf{e}_1 = -\mathbf{e}_3, \qquad \mathbf{e}_3 \times \mathbf{e}_2 = -\mathbf{e}_1, \qquad \mathbf{e}_1 \times \mathbf{e}_3 = -\mathbf{e}_2,$$

with $\mathbf{e}_i \times \mathbf{e}_i = \mathbf{0}$.

Equivalently, after defining the permutation symbol in Class 01,

$$[\mathbf{a} \times \mathbf{b}]_k = \epsilon_{ijk} a_i b_j .$$

Here k is free, while i and j are dummy indices.

Result 1.3 (Skew Tensors and Axial Vectors)

Given any skew-symmetric tensor $W \in \mathcal{V}^2$, there exists a unique vector $\mathbf{w} \in \mathcal{V}$ such that

$$W\mathbf{v} = \mathbf{w} \times \mathbf{v}, \qquad \mathbf{v} \in \mathcal{V}.$$

In any right-handed orthonormal frame $\mathcal{F} = \{\mathbf{e}_i\}$,

$$w_j = [\mathbf{w}]_j = \frac{1}{2}\epsilon_{njm} W_{nm}.$$

Conversely, for each $\mathbf{w} \in \mathcal{V}$, there exists a unique skew-symmetric tensor $W \in \mathcal{V}^2$ such that $W\mathbf{v} = \mathbf{w} \times \mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$, and

$$W_{ik} = \epsilon_{ijk} w_j.$$

Proof.

Fix a right-handed orthonormal frame $\mathcal{F} = \{\mathbf{e}_i\}$. We want to show that, in this frame,

$$W_{ik} = \epsilon_{ijk} w_j. \tag{1.8}$$

For existence, we only need to show that what was proposed above gives the desired result:

$$w_j := \frac{1}{2}\epsilon_{njm} W_{nm}, \qquad j = 1, 2, 3. \tag{1.9}$$

Recall that, once we know the components of a vector in some frame, we can write

$$\mathbf{w} = w_1\mathbf{e}_1 + w_2\mathbf{e}_2 + w_3\mathbf{e}_3.$$

Proof (Cont.)

Using Result 1.1 and equation (1.9),

$$\begin{aligned}
\epsilon_{ijk} w_j &= \epsilon_{ijk} \left(\frac{1}{2} \epsilon_{njm} W_{nm} \right) \\
&= \frac{1}{2} \epsilon_{ijk} \epsilon_{njm} W_{nm} \\
&= \frac{1}{2} (\delta_{in}\delta_{km} - \delta_{im}\delta_{kn}) W_{nm} \\
&= \frac{1}{2} (W_{ik} - W_{ki}) \\
&= W_{ik},
\end{aligned}$$

because $W_{ki} = -W_{ik}$. Thus equation (1.8) holds.

Proof (Cont.)

Let $\mathbf{v} = v_k \mathbf{e}_k$ be arbitrary. Equation (1.8) gives

$$\begin{aligned}[W\mathbf{v}]_i &= W_{ik} v_k \\ &= \epsilon_{ijk} w_j v_k \\ &= [\mathbf{w} \times \mathbf{v}]_i.\end{aligned}$$

Here i is free, while j and k are dummy indices. Therefore

$$W\mathbf{v} = \mathbf{w} \times \mathbf{v} \qquad \text{for every } \mathbf{v} \in \mathcal{V}.$$

Proof (Cont.)

For uniqueness, suppose $\mathbf{w}, \mathbf{u} \in \mathcal{V}$ both satisfy equation (1.8). Then

$$\epsilon_{ijk}(w_j - u_j) = 0, \qquad 1 \leq i, k \leq 3.$$

The nonzero cases $ijk = 123, 231, 312$ give, respectively,

$$\begin{aligned} w_2 - u_2 &= 0, \\ w_3 - u_3 &= 0, \\ w_1 - u_1 &= 0. \end{aligned}$$

Thus $\mathbf{w} = \mathbf{u}$.

Proof (Cont.)

Conversely, let $\mathbf{w} \in \mathcal{V}$ be given and define

$$W_{ik} := \epsilon_{ijk} w_j .$$

The components determine a unique tensor $W \in \mathcal{V}^2$. Moreover,

$$W_{ki} = \epsilon_{kji} w_j = -\epsilon_{ijk} w_j = -W_{ik},$$

after consistently renaming the dummy index, so $W^T = -W$.

As before, for arbitrary $\mathbf{v} = v_k \mathbf{e}_k$,

$$[W\mathbf{v}]_i = W_{ik} v_k = \epsilon_{ijk} w_j v_k = [\mathbf{w} \times \mathbf{v}]_i .$$

The uniqueness of W follows because its components are fixed by (1.8). □

Example (From an Axial Vector to a Skew Tensor)

In a right-handed orthonormal frame, let

$$[\mathbf{w}] = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}.$$

Using $W_{ik} = \epsilon_{ijk} w_j$, we obtain

$$[W] = \begin{bmatrix} 0 & -w_3 & w_2 \\ w_3 & 0 & -w_1 \\ -w_2 & w_1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -3 & -2 \\ 3 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix}.$$

Here i, k are free in the component formula, while j is dummy.

Example (From an Axial Vector to a Skew Tensor (Cont.))

Take

$$[\mathbf{v}] = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}.$$

Matrix multiplication gives

$$[W\mathbf{v}] = \begin{bmatrix} 0 & -3 & -2 \\ 3 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 7 \\ 5 \end{bmatrix}.$$

The cross-product calculation gives the same components:

$$[\mathbf{w} \times \mathbf{v}] = \begin{bmatrix} w_2 v_3 - w_3 v_2 \\ w_3 v_1 - w_1 v_3 \\ w_1 v_2 - w_2 v_1 \end{bmatrix} = \begin{bmatrix} -1 \\ 7 \\ 5 \end{bmatrix}.$$

Hence $W\mathbf{v} = \mathbf{w} \times \mathbf{v}$ in this example.

Symmetric and Skew Parts; Axial Correspondence

Definition

For $S \in \mathcal{V}^2$, define

$$\mathrm{sym}(S) = \frac{1}{2}(S + S^T), \qquad \mathrm{skw}(S) = \frac{1}{2}(S - S^T).$$

Proposition

If $W \in \mathcal{V}^2$ is skew-symmetric, define $\mathrm{vec}(W)$ *as the unique axial vector supplied by Result 1.3. If* $\mathbf{w} \in \mathcal{V}$, *define* $\mathrm{ten}(\mathbf{w})$ *as the unique skew-symmetric tensor supplied by the converse statement. The functions*

$$\mathbf{w} = \mathrm{vec}(W) \qquad \textit{and} \qquad W = \mathrm{ten}(\mathbf{w})$$

are inverses.

Proof.

This follows directly from the two uniqueness statements in Result 1.3. $\square$

Change of Basis

The Change-of-Basis Tensor

Definition

Let

$$\mathcal{F} = \{\mathbf{e}_i\} \qquad \text{and} \qquad \mathcal{F}' = \{\mathbf{e}_i'\}$$

be two right-handed orthonormal frames for $\mathcal{V}$. The *change-of-basis tensor* from $\mathcal{F}'$ to $\mathcal{F}$ is

$$A = A_{ij}\, \mathbf{e}_i \otimes \mathbf{e}_j, \qquad A_{ij} = \mathbf{e}_i \cdot \mathbf{e}_j'. \tag{1.11}$$

Proposition

For the two frames $\mathcal{F} = \{\mathbf{e}_i\}$ *and* $\mathcal{F}' = \{\mathbf{e}'_i\}$,

$$\mathbf{e}'_j = A_{ij}\mathbf{e}_i, \qquad 1 \le j \le 3, \tag{1.12}$$

and, similarly,

$$\mathbf{e}_i = A_{ik}\mathbf{e}'_k, \qquad 1 \le i \le 3. \tag{1.13}$$

Proof.

Because $\mathcal{F}$ is a basis, write

$$\mathbf{e}'_j = c_1\mathbf{e}_1 + c_2\mathbf{e}_2 + c_3\mathbf{e}_3.$$

Orthonormality gives

$$\mathbf{e}'_j \cdot \mathbf{e}_k = c_k.$$

Therefore,

$$\begin{aligned}
\mathbf{e}'_j &= (\mathbf{e}'_j \cdot \mathbf{e}_1)\mathbf{e}_1 + (\mathbf{e}'_j \cdot \mathbf{e}_2)\mathbf{e}_2 + (\mathbf{e}'_j \cdot \mathbf{e}_3)\mathbf{e}_3 \\
&= (\mathbf{e}'_j \cdot \mathbf{e}_k)\mathbf{e}_k \\
&= (\mathbf{e}_k \cdot \mathbf{e}'_j)\mathbf{e}_k \\
&= A_{kj}\mathbf{e}_k,
\end{aligned}$$

which is (1.12).

Proof (Cont.)

Next, expanding in the primed frame,

$$\begin{aligned} \mathbf{e}_i &= (\mathbf{e}_i \cdot \mathbf{e}'_k)\mathbf{e}'_k \\ &= A_{ik}\mathbf{e}'_k, \end{aligned}$$

which is (1.13). □

Result 1.4 (Orthogonality of the Change-of-Basis Matrix)

The components A_{ij} of the change-of-basis tensor satisfy

$$A_{ij}A_{ik} = \delta_{jk} \qquad \text{and} \qquad A_{ik}A_{\ell k} = \delta_{i\ell}.$$

In matrix notation,

$$[A]^T[A] = [I], \qquad [A][A]^T = [I].$$

Thus the change-of-basis matrix is orthogonal.

Proof.

Substituting (1.13) into (1.12),

$$\mathbf{e}'_j = A_{ij} A_{ik} \mathbf{e}'_k.$$

But also $\mathbf{e}'_j = \delta_{jk} \mathbf{e}'_k$. Uniqueness of components gives

$$A_{ij} A_{ik} = \delta_{jk}.$$

Similarly, substituting (1.12) into (1.13) yields

$$\mathbf{e}_i = A_{ik} A_{\ell k} \mathbf{e}_\ell,$$

and hence

$$A_{ik} A_{\ell k} = \delta_{i\ell}.$$

□

Vector Components under a Change of Basis

Proposition

Let $\mathbf{v} \in \mathcal{V}$ *be arbitrary, with*

$$[\mathbf{v}]_{\mathcal{F}} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}, \qquad [\mathbf{v}]_{\mathcal{F}'} = \begin{bmatrix} v_1' \\ v_2' \\ v_3' \end{bmatrix}.$$

If A *is the change-of-basis tensor from* $\mathcal{F}'$ *to* $\mathcal{F}$, *then*

$$v_i = A_{ij} v_j', \qquad v_k' = A_{ik} v_i.$$

Equivalently,

$$[\mathbf{v}]_{\mathcal{F}} = [A]_{\mathcal{F}} [\mathbf{v}]_{\mathcal{F}'}, \qquad [\mathbf{v}]_{\mathcal{F}'} = [A]_{\mathcal{F}}^{T} [\mathbf{v}]_{\mathcal{F}}.$$

Proof.

Using $\mathbf{v} = v'_j \mathbf{e}'_j$ and equation (1.12),

$$\mathbf{v} = v'_j A_{ij} \mathbf{e}_i .$$

Comparison with $\mathbf{v} = v_i \mathbf{e}_i$ gives

$$v_i = A_{ij} v'_j .$$

Using $\mathbf{v} = v_i \mathbf{e}_i$ and equation (1.13),

$$\mathbf{v} = v_i A_{ik} \mathbf{e}'_k .$$

Comparison with $\mathbf{v} = v'_k \mathbf{e}'_k$ gives

$$v'_k = A_{ik} v_i .$$

□

A Compact Notational Convention

When the frames are understood, we write

$$[\,\cdot\,] = [\,\cdot\,]_{\mathcal{F}} \qquad \text{and} \qquad [\,\cdot\,]' = [\,\cdot\,]_{\mathcal{F}'}.$$

Thus, for example,

$$[\mathbf{v}] = [A][\mathbf{v}]', \qquad [\mathbf{v}]' = [A]^T[\mathbf{v}].$$

Tensor Components under a Change of Basis

Proposition

Suppose $S \in \mathcal{V}^2$ is arbitrary and A is the change-of-basis tensor from $\mathcal{F}'$ to $\mathcal{F}$. Then

$$[S] = [A][S]'[A]^T$$

and

$$[S]' = [A]^T[S][A].$$

Proof.

Exercise. $\square$

Traces, Determinants, and Exponentials

Trace

Definition

Let $S \in \mathcal{V}^2$ be arbitrary and suppose $\mathcal{F} = \{\mathbf{e}_i\}$ is a frame. Define

$$\mathrm{tr}(S) = \mathrm{tr}([S]) = S_{ii}.$$

Here $[S]$ is the matrix representation of S in $\mathcal{F}$.

At first, this definition appears to depend on the chosen frame. Result 1.5 shows that it does not.

Result 1.5 (Invariance of Trace)

Let $S \in \mathcal{V}^2$ be given, with $[S]$ and $[S]'$ its representations in two frames $\mathcal{F}$ and $\mathcal{F}'$. Then

$$\mathrm{tr}([S]) = \mathrm{tr}([S]').$$

Proof.

Using the change-of-basis formula,

$$\begin{aligned}
\mathrm{tr}([S]) &= S_{ii} \\
&= A_{ik} S'_{k\ell} A_{i\ell} \\
&= S'_{k\ell} A_{ik} A_{i\ell} \\
&= S'_{k\ell} \delta_{k\ell} \\
&= S'_{kk} \\
&= \mathrm{tr}([S]').
\end{aligned}$$

$\square$

Determinant

Definition

Let $S \in \mathcal{V}^2$ and suppose $\mathcal{F} = \{\mathbf{e}_i\}$ is a frame. Define

$$\det(S) = \det([S]) = \epsilon_{ijk}[S]_{i1}[S]_{j2}[S]_{k3}.$$

Result 1.6 (Invariance of Determinant)

If $[S]$ and $[S]'$ represent S in two frames, then

$$\det([S]) = \det([S]').$$

Proof.

Exercise. □

Basic Determinant Properties

Proposition

Let $S, T \in \mathcal{V}^2$ be arbitrary. Then S is invertible if and only if

$$\det(S) \neq 0.$$

Moreover,

$$\det(ST) = \det(S)\det(T) = \det(TS).$$

If $Q \in \mathcal{V}^2$ is orthogonal, then

$$|\det(Q)| = 1.$$

If Q is a rotation, then

$$\det(Q) = 1.$$

Tensor Exponential

Definition

For $S \in \mathcal{V}^2$, define

$$\exp(S) = I + S + \frac{1}{2}S^2 + \frac{1}{6}S^3 + \cdots = \sum_{j=0}^{\infty} \frac{1}{j!}S^j.$$

This series converges for every $S \in \mathcal{V}^2$.

Result 1.7 (Exponentials of Skew Tensors)

Let $W \in \mathcal{V}^2$ be skew-symmetric. Then $\exp(W)$ is a rotation tensor. In particular,

$$(\exp(W))^T \exp(W) = I$$

and

$$\det(\exp(W)) = 1.$$

Furthermore,

$$\exp(W)^T = \exp(W^T).$$

Proof.

Exercise. $\square$